

Mata

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**A comprehensive guide to
the language**

Written in English

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Mata

0 Introduction

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Welcome to this comprehensive guide to my language known as *Mata*.

This is my first conlang, made to revolve around mathematical notation, in particular set notation. The core concept to know is that almost everything can be described as a set, containing all of the information about said thing. For example, a person can be stated to be a set (i.e. $\mathbb{P}$) that contains all of the adjectives describing them. We can then make statements about that set (i.e. " $\mathbb{P} \ni \text{stupid}$ " *aka* the person is stupid, however "stupid" wouldn't just be in English).

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1 Lexicon

1.1 Alphabet

Using the IPA for the sounds

Symbol	Sound	Symbol	Sound	Symbol	Sound	Symbol	Sound	Symbol	Sound
Aa	æ	Gg	g	Mm	m	Ss	s	Yy	j
Bb	b	Hh	h	Nn	n	Tt	t	Zz	z
Cc	tʃ	Ii	ɪ	Oo	ɒ	Uu	ʊ		
Dd	d	Jj	ʒ	Pp	p	Vv	v		
Ee	ɛ	Kk	k	Qq	θ	Ww	w		
Ff	f	Ll	l	Rr	r	Xx	ʃ		

For other sounds, the symbol (,) can be used to modify certain symbols. For vowels it converts them into their long forms
(Written in L^AT_EX as ”_\prime”)

List of alternate sounds:

Symbol	Sound
<i>a,</i>	eɪ
<i>e,</i>	i:
<i>i,</i>	aɪ
<i>o,</i>	oʊ
<i>u,</i>	u:
<i>q,</i>	ð

”*q,*” is very uncommon but does exist, primarily for words taken from other languages being written in Mata’s alphabet. For example, ”The” would be written as ”*q^h*”. The other symbols in this will be touched on shortly.

Mata very rarely uses the normal symbols for vowels outside of the beginning of words, or pluralisation. Instead, Mata uses diacritics to denote when a vowel follows a consonant. A table of the diacritic marks follows:

Symbol	Equivalent vowel
ô	a
ö	e
õ	i
ó	o
ü	u

Additionally there are two other diacritics other than for vowels.
To asperate a sound, a superscript 'h' can be appended.
To extend a sound, a bar can be added above or below the symbol. If the bar is above another diacritic, it will extend the sound of the diacritic rather than the base consonant.

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1.2 Special Characters

Mata uses more than a basic alphabet. There are a few special characters with dedicated meanings. All special characters from now on will have their sounds written in the Mata alphabet.

Symbol	L ^A T _E X	Sound	Meaning
$\mathcal{E}$	<code>\mathcal{E}</code>	<i>e</i> ,	The speaker (I)
$\mathfrak{U}$	<code>\mho</code>	<i>û</i>	The listener (you singular)
$\emptyset$	<code>\emptyset</code>	<i>ĥ</i>	0 / nothing
ξ	<code>\xi</code>	<i>a^h</i>	Everything
σ	<code>\sigma</code>	<i>ṣ̌, g</i>	Level of significance

1.3 Operators

Operators are special characters which describe properties or modifications to lexical objects. In sentences, they are particles that fit between objects.

Symbol	L ^A T _E X	Sound	Meaning
$+$	<code>+</code>	<i>p̣ḷs</i>	Addition
$-$	<code>-</code>	<i>ṃñs</i>	Subtraction
$\times$	<code>\times</code>	<i>ṃḷṭ</i>	Multiplication
$\div$	<code>\div</code>	<i>ḍṿj</i>	Division
$\ni / \in$	<code>\ni / \in</code>	<i>ḳ</i> or <i>ḳ,</i>	Contains / Is contained in
$\not\ni / \notin$	<code>\not\ni / \notin</code>	<i>ṇ</i> $\ni$ or <i>ṇ,</i> $\ni$	Does not contain / is not contained in
$\circ$	<code>\circ</code>	<i>k</i> (short)	Used to access an object's child
$\cdot$	<code>\cdot</code>	<i>also k</i> (short)	Decimal place
$\neg$	<code>\neg</code> (often on keyboards)	<i>ṇ</i> or <i>ṇ,</i>	Not, for saying something is false
$\dashv$	<code>\dashv</code>	<i>ṃ</i> or <i>ṃ,</i>	Not, for inverting
$\vdash$	<code>\vdash</code>	<i>oyl</i>	Was true and is still true.
$\updownarrow$	<code>\updownarrow</code>	<i>vr</i>	Vertical
$\leftrightarrow$	<code>\leftrightarrow</code>	<i>hr</i>	Horizontal
$=$	<code>=</code>	<i>a^h</i>	is/equals
$\neq$	<code>\neq</code>	<i>na^h</i>	is not/not equal
$\overset{*}{=}$	<code>\overset{*}{=}</code>	<i>oŕ</i>	Requires/depends on
$\supset / \subset$	<code>\supset / \subset</code>	<i>ṣ</i> or <i>ṣ,</i>	Is a subset of / contains the set... (plural of $\in$)
$\not\supset / \not\subset$	<code>\not\supset / \not\subset</code>	<i>ṇ</i> $\supset$ or <i>ṇ,</i> $\supset$	negative of subset/superset
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Symbol	L ^A T _E X	Sound	Meaning
?	?	<i>br</i>	Convert an operator to a question
{ / }	\{ / \}	<i>î</i>	Open/Close a set. "}" does not have to be pronounced at the end of a sentence.
		<i>Ǧ</i>	And, in a set
∩	\cap	<i>Ǧ,y</i>	And, between sets (intersection)
∪	\cup	<i>Ǧy</i>	Or, between sets (union)
∴	\therefore	<i>ǦǦ,</i>	Therefore
∵	\because	<i>ǦǦ,</i>	Because
◊	\diamond	<i>Ǧ</i>	Is possible (can)
⊞	\boxplus	<i>ǦpǦ</i>	However/But (Despite this)
∧	\land	<i>Ǧ</i>	And, in logic
∨	\vee	<i>Ǧ</i>	Or, in logic
:	\colon	No sound	Used optionally to separate adjacent words as Mata does not use spaces

1.4 Numbers

Mata, like other western languages, uses base 10 and the same numerals as English (Hindu-Arabic numerals if i remember correctly). The names for numbers are as follows:

Numeral	Word	Numeral	Word	Numeral	Word
0	<i>na</i>	10	<i>ǧ,n</i>	100	<i>ǧ,t</i>
1	<i>ak</i>	11	<i>ǧ,nak</i>	200	<i>ǧ,t</i>
2	<i>el</i>	20	<i>ǧ,n</i>	300	<i>ǧ,t</i>
3	<i>dǦ</i>	30	<i>ǧ,n</i>	1000	<i>ǧ,y</i>
4	<i>fǧ</i>	40	<i>ǧ,n</i>	2000	<i>ǧ,y</i>
5	<i>ǧn</i>	50	<i>ǧ,n</i>	3000	<i>ǧ,y</i>
6	<i>ǧn</i>	60	<i>ǧ,n</i>		
7	<i>pǦ</i>	70	<i>pǦ,n</i>		
8	<i>uv</i>	80	<i>ǧ,n</i>		
9	<i>ǧc</i>	90	<i>ǧ,n</i>		

If you wish to say a number with order above 3, you can use scientific notation with the *p* particle (i.e 2.5 million = 2.5×10^6 = *elǧǧnǧǧn*).

Numbers can be used for emphasis like an adjective (i.e. "very good" would be like saying "10 good"/"*ǧ,n : ǧnl*").

The ordinal variants are as follows:

Number	1	2	3	4	5
Word	<i>ǧ,</i>	<i>ǧ</i>	<i>dǦ</i>	<i>fǦ</i>	<i>ǧn</i>

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2 Words

Time for an overview of some of the words in *Mata*. These lists are not extensive and at somepoint I will create a dictionary for them.

2.1 Verbs

Verbs is not the best word to describe these words, they more represent the concept of the action rather than the action itself. This will be more relevant later. Here is a list of some of the more common *Verbs* (or some of the first few I created) with rough English translations:

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
To do	$\dot{d}$	To say/speak	$\hat{S}g$	To play	$\acute{s}nd$	To hate	$\hat{d}\ddot{r}$	To fish	$\hat{s}\hat{m}$
To read	$k\ddot{o}c$	To go	$\check{m}ln$	To consume	$v\acute{l},s$	To get	$\acute{r}\hat{r}$	To know	$o\check{y}lr$
To want	$\check{y}tx$	To see	$b\check{y}n$	To sex	$\check{l}\hat{f}$	To assume/ think/ believe	$\hat{d}rs$	To milk	$\check{m}lj$
To be going to/intend to (will)	$\check{q}\hat{v}t$	To assert (suggest is true)	$\check{k}l\hat{t}$	To die/kill	$ol\hat{k}$				

Mata supports compound *verbs* as well:

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
To sleep	$n\check{y}c\check{m}ln$	To drive	$\hat{m}\check{x}n\check{m}ln$	To love	$\neg \hat{d}\ddot{r}$	To hunger/ thirst	$v\acute{l},s\check{y}tx$		

In order to describe the past or future tense of a *verb*, you can either

- Access the past or future child of the *verb* (i.e. $\mathcal{E} \circ \dot{d} \circ \tilde{u}p$ is the set of things that I, the speaker, have done)
- Apply the past or future as a function, less uses (i.e. $\tilde{u}p(\hat{s}\hat{m})$ is the state of having fished, The sentence "he is going to have fished" would be $\tilde{G} \circ \dot{d} \circ k\check{l} \ni \tilde{u}p(\hat{s}\hat{m}))$

Here are a few example sentences:

<i>James says "Hello"</i>	$D\hat{j},mz \circ \hat{S}g \ni "$	$\check{H}\check{l},"$	(James'k sag koh Hello)
<i>James is saying "Hello"</i>	$D\hat{j},mz \circ \dot{d} \ni \hat{S}g("$	$\check{H}\check{l},"$)	(James'k doh koh sag Hello)
<i>James is doing nothing</i>	$D\hat{j},mz \circ \dot{d} = \emptyset$		(James'k doh ah na)
<i>I hate you</i>	$\mathcal{E} \circ \hat{d}\ddot{r} \ni \mathcal{U}$		(Ee'k dareh koh wa)
<i>I hate you a lot</i>	$\mathcal{E} \circ \hat{d}\ddot{r} : \check{k},\hat{y} \ni \mathcal{U}$		(Ee'k dareh keen koh wa)
<i>I really really love you</i>	$\mathcal{E} \circ \neg \hat{d}\ddot{r} : \check{k},\hat{y} \ni \mathcal{U}$		(Ee'k modareh keeya koh wa)
<i>I saw you</i>	$\mathcal{E} \circ b\check{y}n \circ \tilde{u}p \ni \mathcal{U}$		(Ee'k byan k'u-ip koh wa)
<i>I saw you reading</i>	$\mathcal{E} \circ b\check{y}n \circ \tilde{u}p \ni \mathcal{U} \circ \dot{d}(k\ddot{o}c)$		(Ee'k byan k'u-ip koh wa'k doh(ko-ich))
<i>James can't do everything</i>	$D\hat{j},mz \circ \diamond \dot{d} \neq \xi$		(James k'padoh na ah)
<i>I want to go to sleep</i>	$\mathcal{E} \circ \check{y}tx \ni n\check{y}c\check{m}ln$		(Ee'k yetsh koh nyoch-muln)

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2.2 Nouns and Pronouns

2.2.1 Pronouns

Mata supports most of the same pronouns as *English* but remember that objects like pronouns are sets which can be combined together to describe different groups.

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
I	$\mathcal{E}$	You	$\mathcal{U}$	they/it	$\tilde{g}$	You(p) [†]	$p\hat{y}$	they(p) [‡]	$g\hat{y}$
This/ These	$\mathcal{E}\hat{c}$	That/ Those	$\mathcal{U}\hat{c}$						

[†]Any group of people including the listener and not including the speaker

[‡]Any group of people including neither the listener nor the speaker

2.2.2 Nouns

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
Past	$\tilde{u}p$	Future	$k\check{l}$	Name	$\hat{n},m$	Night	$n\acute{y}c$	Apple	$a\check{p}l$